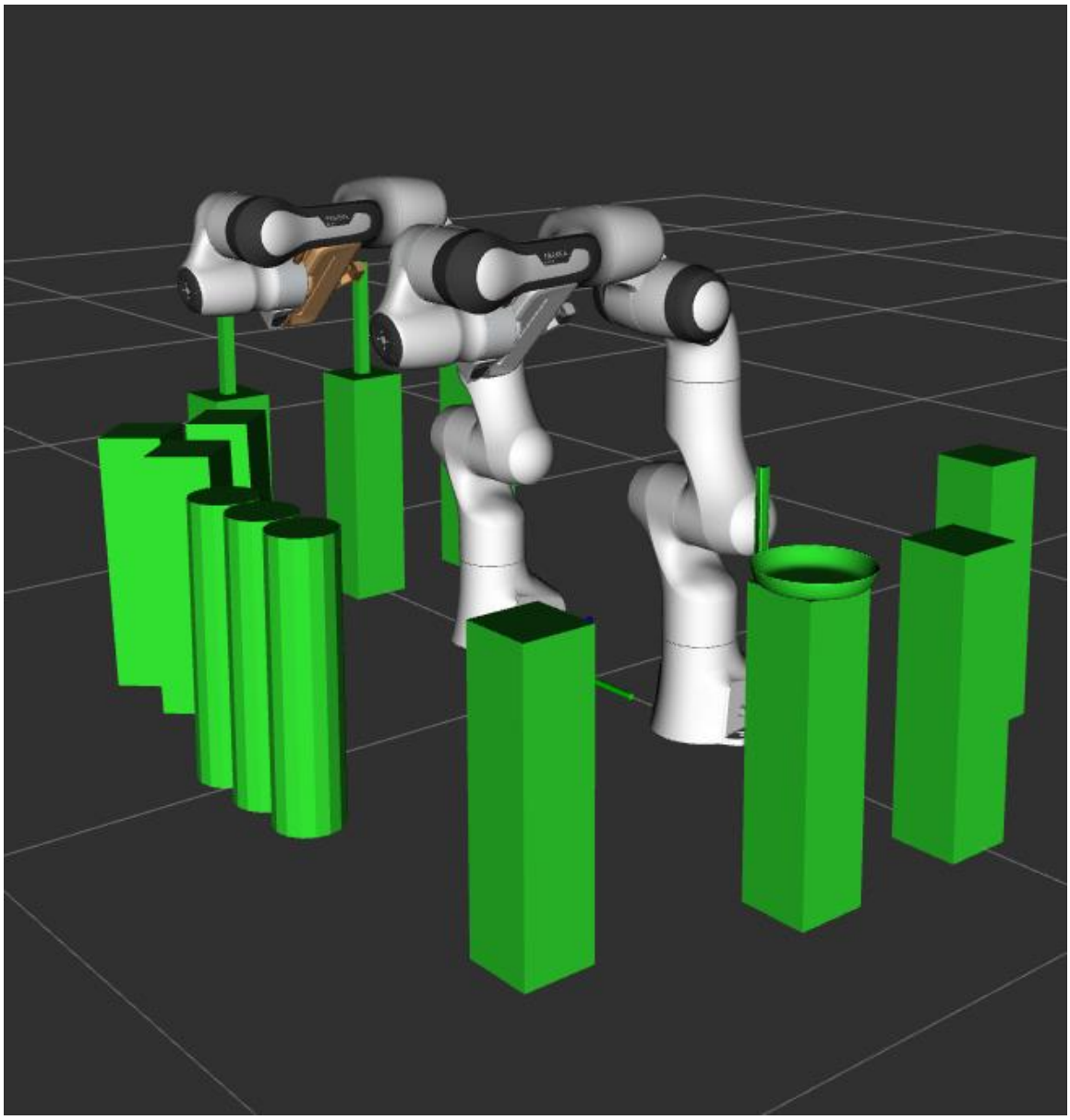


Collaborative Dual Arm Robot

PERCEPTION

Philopatear Ashraf Salib| June 2020

Abstract

Robots are becoming increasingly more popular and present in our lives; we find them in factories and even in our homes as cleaning bots. Introducing them to new environments means being able to help us more, here we introduce them to the kitchen environment.

The project consists of two Franka Emika dual collaborative arms working together to cook pasta. The whole system consists of task planning, motion planning and perception module. Orders are received by the task-planning module, divided into subtasks and sent to the motion-planning module for execution within the given environment, while the perception module informs the motion planning about different items' positions in the environment.

The main purpose of the perception module as a primary step in this report is to construct a system that detects, identifies and localizes different items in an environment. Once an appropriate dataset of an environment is annotated, added to the system for training, system can detect, identify and localize a given item and communicate this information to other modules in the system.

This was done by PyTorch library using a VGG-16 base pretrained on ImageNet that is already available in PyTorch's torchvision module. VOC 2007 and 2012 was used as training and testing dataset. Also KITTI dataset was used to prove that this classifier can be used in various applications including autonomous vehicles.

The perception module has proven that once an appropriate annotated dataset is provided and with the use of a Depth Camera, the perception module will be able to provide an accurate information of different items in an environment.

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Introduction

Autonomous industry is being developed quite fast and has shown a great impact on everyday life in the past few years, accordingly automation has been recently presented to many unpopular places for robots such as in this case, kitchens.

This project consists of two Franka Emika dual collaborative arms working together to cook pasta. The system consists of task planning, motion planning and perception module. Orders are received by the task-planning module, divided into subtasks and sent to the motion-planning module for execution within the given environment, while the perception module informs the motion planning about different items' positions in the environment.

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Problem Definition

The project consists of two Franka Emika arms working together to perform a series of tasks with the end goal of serving an order of pasta. The environment that the two robots are working in is a dynamic environment with lots of potential mishaps. This is where the perception module comes in.

In order to avoid such mishaps, the objects in the system needs to be identified and localized within the environment and that knowledge to be shared with the motion-planning module as a feedback for each task's accomplishment.

According to the environment configured; pans, strainers, robots, pasta and sauce dispensers and, cooking, boiling, dirty pan, and clean pan stations are required to be detected, identified, and localized.

Implementation

CLASSIFIER CONSTRUCTION

The Classifier uses PyTorch library, after going through ORK, YOLO, TensorFlow, Keras and OpenCV, PyTorch was chosen for its:

- Versatility
- Flexibility
- Better modular design
- Supports dynamic computational graphs
- Easy debugging
- Easier to Start with

The objective is to build a model that can detect and localize objects in video stream. For this task, Single Shot Multibox Detector (SSD) is implemented. SSD is a popular algorithm in object detection. It is generally faster than Faster RCNN.

Earlier object detection architectures consisted of two separate stages: A regional proposal network for object localization and A Classifier to detect object.

These can be very expensive computationally, thus unfit for real world and real-time applications.

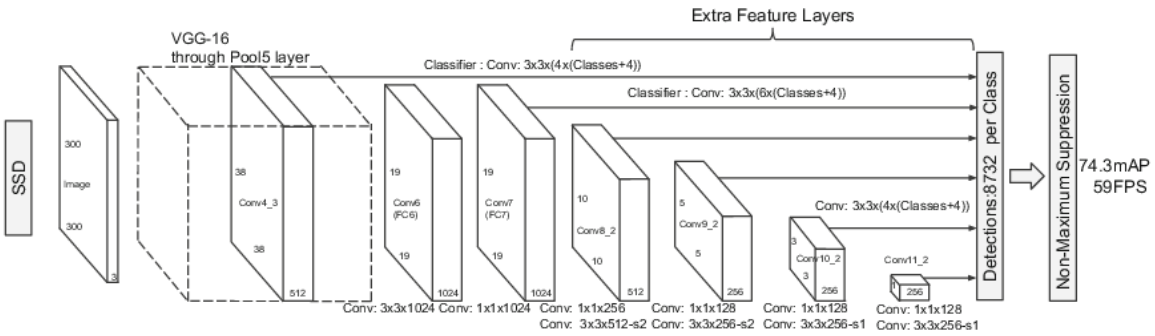


Figure 1: SSD Configuration

On the other hand, as shown in figure 1 Single-shot models encapsulate localization and detection in a single forward sweep of the network. Hence, being more efficient.

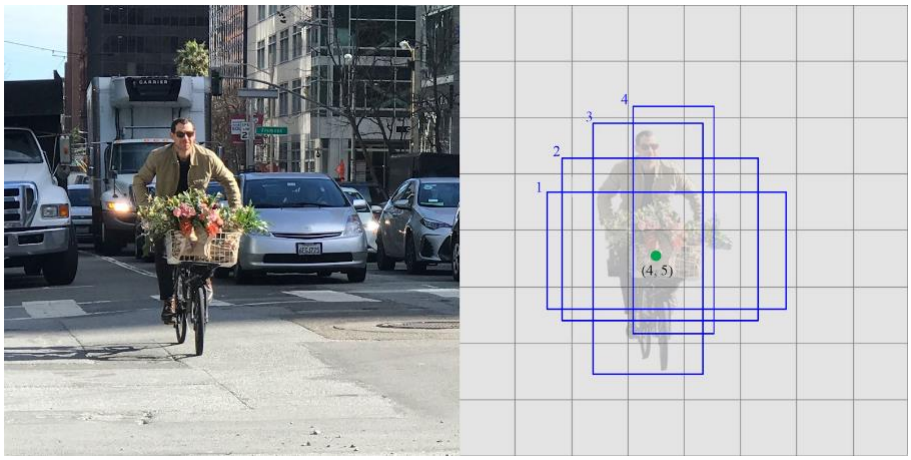


Figure 2: Single Shot Multibox Detector (SSD)

SSD uses non-maximum suppression to remove duplicate predictions pointing to the same object. Non-maximum suppression is a key post-processing step in many computer vision applications. SSD sorts the predictions by the confidence scores. Start from the top confidence prediction, SSD evaluates whether any previously predicted boundary boxes have an IoU higher than 0.45 with the current prediction for the same class. If found, the current prediction will be ignored. At most, we keep the top 200 predictions per image.

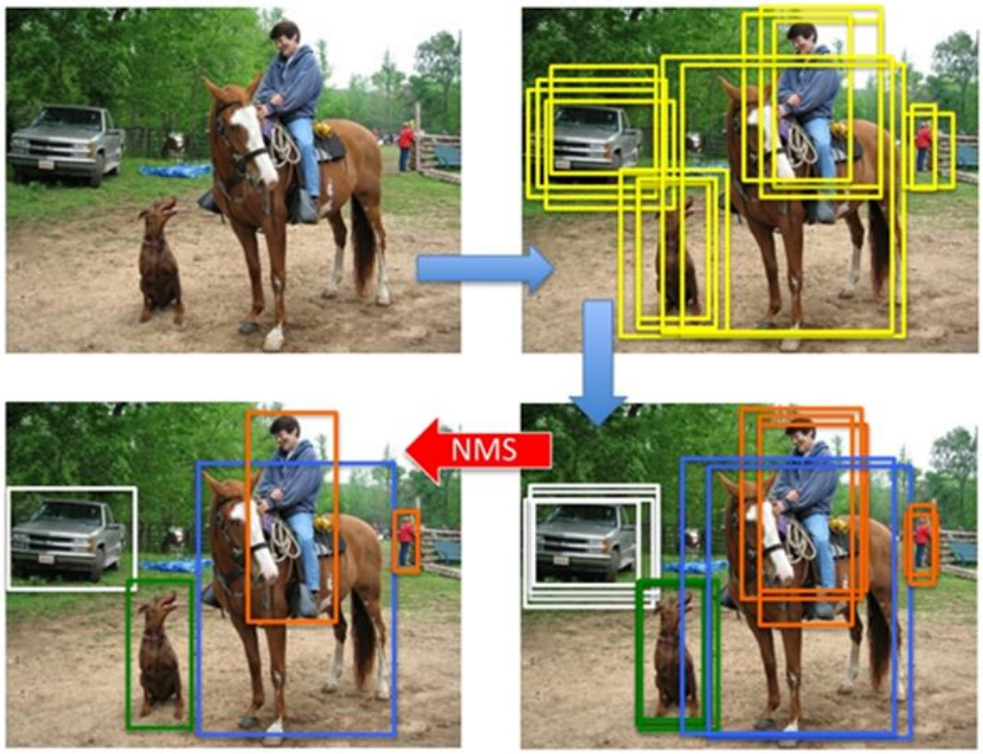


Figure 3: non-maximum suppression

Integration Node

As a proof for the concept of integration with the other modules in the system, a ROS node was created to publish "/coordinates" topic. Each message has a label name and x,y,z coordinates relative to the image.

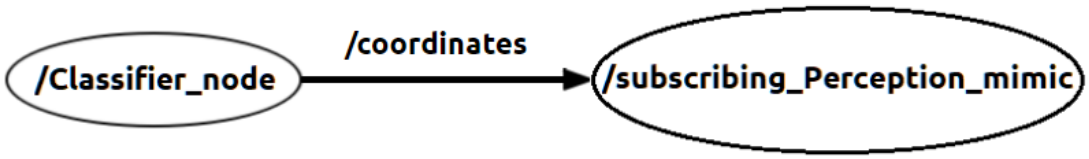


Figure 4: RQT graph of integration between perception and motion-planning module

Results

Using the VOC 2007 & VOC 2012 Testing Dataset, the following accuracies have been achieved:

Class	Average Precision	Class	Average Precision
aeroplane	0.788	*dog*	0.856
bicycle	0.835	*horse*	0.878
bird	0.762	*motorbike*	0.832
boat	0.721	*person*	0.788
bottle	0.459	*pottedplant*	0.507
bus	0.870	*sheep*	0.794
car	0.865	*sofa*	0.8
cat	0.883	*train*	0.866
chair	0.592	*tvmonitor*	0.749
cow	0.826	*diningtable*	0.757

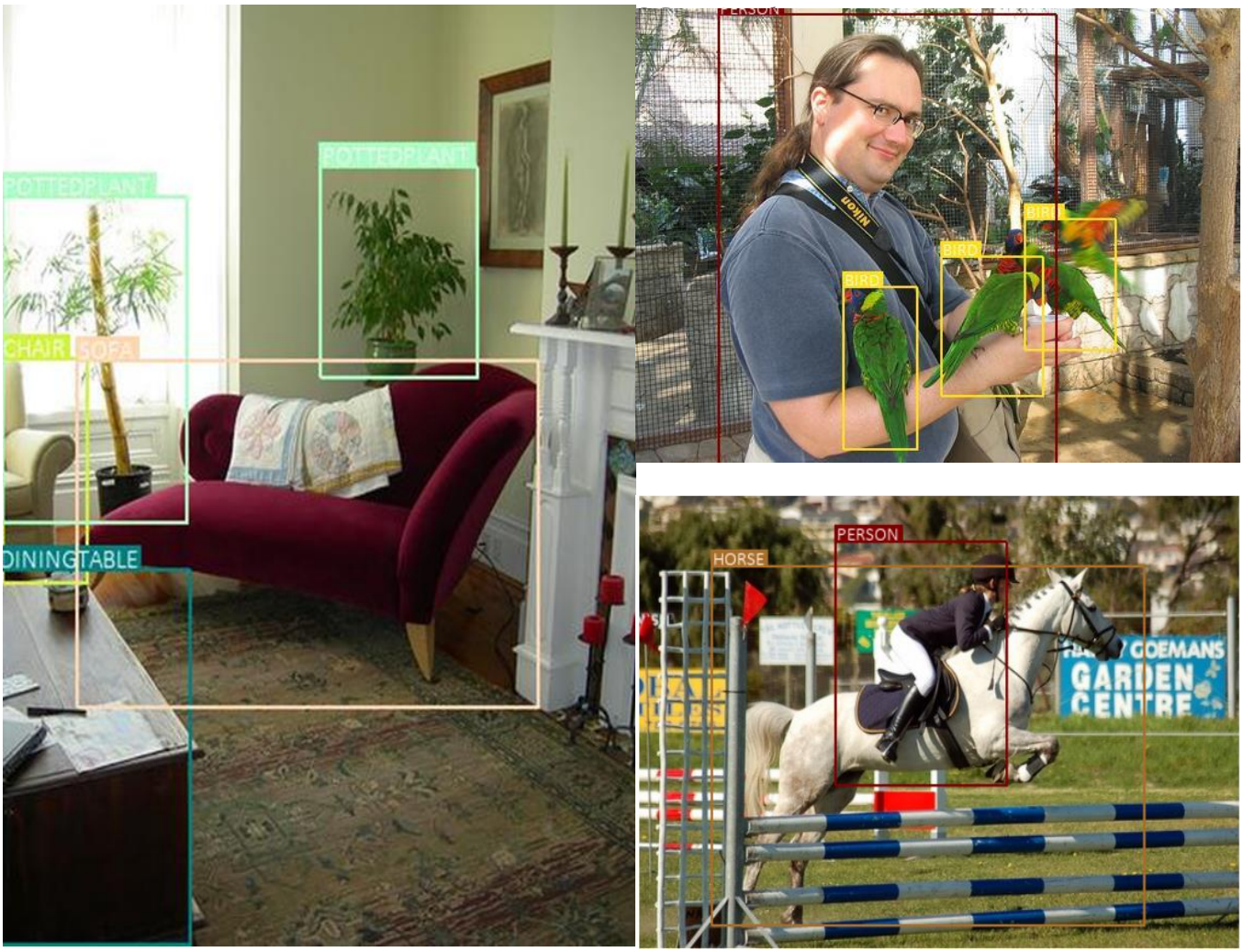


Figure 5: Results of VOC dataset

The KITTI dataset has been used to show that this classifier can work with different datasets including autonomous vehicles; the classifier can identify and localize objects like cars, person, trains ... etc. The KITTI dataset has been recorded from a moving platform while driving in and around Karlsruhe, Germany. It includes camera images, laser scans, high-precision GPS measurements and IMU accelerations from a combined GPS/IMU system.

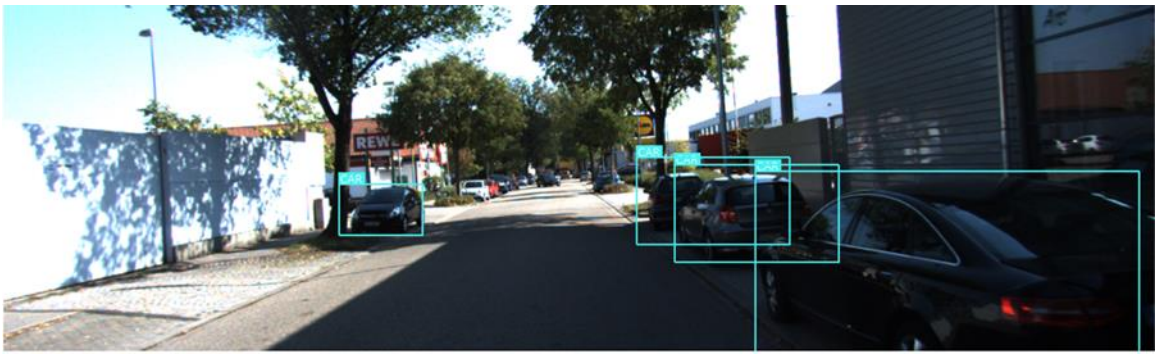


Figure 6: Results of KITTI Dataset

For integration with the motion-planning module, a pre-defined set of coordinates for the clean pan position was published on the coordinates' topic. Accordingly, the motion-planning module picked the clean pan from the clean pan station in compliance to the perception module information provided.

Conclusion

The main purpose of the perception module as a primary step in this report is to construct a system that detects, identifies and localizes different items in an environment. Once an appropriate dataset of an environment is annotated, added to the system for training, system can detect, identify and localize a given item and communicate this information to other modules in the system.

This was done by PyTorch library using a VGG-16 base pre-trained on ImageNet that is already available in PyTorch's torchvision module. VOC 2007 and 2012 was used as training and testing dataset. Single Shot Multibox Detector (SSD) is implemented for object detection, and has used non-maximum suppression to remove duplicate predictions pointing to the same object to remove duplicate predictions pointing to the same object, since SSD to remove duplicate predictions pointing to the same object.

A ROS node has been created to prove the ability of integration between the perception and the motion-planning module.

The classifier is applied to the Kitti Dataset and it was able to identify and localize objects like cars, person, trains … etc.

The perception module has proven that once an appropriate annotated dataset is provided and with the use of a Depth Camera, the perception module will be able to provide an accurate information of different items in an environment.